

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 4: Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Word problems

1. Over a 200 km journey, a fast train takes one hour less than a slow train. The slow train runs 10 km/h slower than the fast one. Find the speed of each train. **CBSE 2026, 5 marks**

SOLUTION

Let the fast train run at x km/h. Then the slow train runs at $(x - 10)$ km/h.

$$\frac{200}{x - 10} - \frac{200}{x} = 1$$

$$200x - 200(x - 10) = x(x - 10), \text{ so } 2000 = x^2 - 10x$$

$$x^2 - 10x - 2000 = 0, \text{ so } (x - 50)(x + 40) = 0$$

$x = 50$ or $x = -40$. A speed cannot be negative.

Check: $200 \div 50 = 4$ hours and $200 \div 40 = 5$ hours, which is one hour more.

Answer: 50 km/h and 40 km/h

2. Two squares have a total area of 640 m^2 , and their perimeters differ by 64 m. Find the side of each square. **CBSE 2026, 5 marks**

SOLUTION

Let the sides be x m and y m with $x > y$.

$$4x - 4y = 64, \text{ so } x = y + 16.$$

$$x^2 + y^2 = 640: (y + 16)^2 + y^2 = 640, \text{ so } 2y^2 + 32y - 384 = 0$$

$$y^2 + 16y - 192 = 0, \text{ so } (y + 24)(y - 8) = 0$$

$y = 8$ (a length cannot be negative) and $x = 24$.

$$\text{Check: } 24^2 + 8^2 = 576 + 64 = 640.$$

Answer: 24 m and 8 m

3. A right-angled triangle has a perimeter of 60 cm and a hypotenuse of 25 cm. Find the lengths of the other two sides. **CBSE 2025, 5 marks**

SOLUTION

Let one of the other sides be x cm. The two sides add up to $60 - 25 = 35$ cm, so the other side is $(35 - x)$ cm.

By the Pythagoras theorem, $x^2 + (35 - x)^2 = 25^2$.

$x^2 + 1225 - 70x + x^2 = 625$, so $2x^2 - 70x + 600 = 0$, that is $x^2 - 35x + 300 = 0$.

$(x - 15)(x - 20) = 0$, so $x = 15$ or $x = 20$.

If $x = 15$ the other side is 20, and if $x = 20$ the other side is 15.

Answer: 15 cm and 20 cm

4. A train covers 480 km at a uniform speed. Had it travelled 8 km/h slower, the journey would have taken 3 hours longer. Find the speed of the train. **CBSE 2025, 5 marks**

SOLUTION

Let the speed be x km/h. The usual time is $\frac{480}{x}$ hours and the slower time is $\frac{480}{x - 8}$ hours.

$$\frac{480}{x - 8} - \frac{480}{x} = 3$$

$480x - 480(x - 8) = 3x(x - 8)$, so $3840 = 3x^2 - 24x$

$x^2 - 8x - 1280 = 0$, so $(x - 40)(x + 32) = 0$

$x = 40$ or $x = -32$. A speed cannot be negative.

Check: $480 \div 40 = 12$ hours and $480 \div 32 = 15$ hours, which is 3 hours more.

Answer: 40 km/h

5. A rectangular floor is covered exactly by 200 square tiles. With tiles whose side is 1 unit longer, only 128 tiles are needed. Let the side of an original tile be x units. (i) Form a quadratic equation in x . (ii) Write it in standard form. (iii) Find x by factorisation, OR solve the equation by the quadratic formula. **CBSE 2024, 4 marks, case study**

SOLUTION

(i) The floor area is the same in both cases: $200x^2 = 128(x + 1)^2$

(ii) $200x^2 = 128x^2 + 256x + 128$, so $72x^2 - 256x - 128 = 0$. Dividing by 8: $9x^2 - 32x - 16 = 0$

(iii) By factorisation: $9x^2 - 36x + 4x - 16 = 0$, so $9x(x - 4) + 4(x - 4) = 0$ and $(9x + 4)(x - 4) = 0$

$x = 4$ or $x = -\frac{4}{9}$. A length cannot be negative, so $x = 4$.

OR by the formula: $x = \frac{32 \pm \sqrt{1024 + 576}}{18} = \frac{32 \pm 40}{18}$, which gives $x = 4$ or $x = -\frac{4}{9}$.

Answer: the side of a tile is 4 units.

6. A school photo is 18 cm long and 12 cm wide. Its length and width are each to be increased by x cm so that the area doubles. (i) Write an equation for this. (ii) Write it as a quadratic equation in standard form. (iii) Find the new length and width, OR decide whether any rational value of x gives a new area of 220 cm². **CBSE 2023, 4 marks**

SOLUTION

(i) Original area = $18 \times 12 = 216$ cm², so the new area is 432 cm². $(18 + x)(12 + x) = 432$

(ii) $216 + 30x + x^2 = 432$. $x^2 + 30x - 216 = 0$

(iii) $(x + 36)(x - 6) = 0$, so $x = 6$ (a length cannot be negative). **New size: 24 cm by 18 cm.**

OR: $(18 + x)(12 + x) = 220$ gives $x^2 + 30x - 4 = 0$, with $D = 900 + 16 = 916$. This is not a perfect square, so the roots are irrational. **No rational value of x works.**

2 Nature of roots

7. The equation $ax^2 + bx + c = 0$, $a \neq 0$, has real and equal roots. Which relation holds? (a) $a = \frac{b^2}{c}$ (b) $b^2 = ac$ (c) $ac = \frac{b^2}{4}$ (d) $c = \frac{b^2}{a}$ **CBSE 2024, 1 mark**

SOLUTION

For real and equal roots, $D = b^2 - 4ac = 0$.

So $b^2 = 4ac$, that is $ac = \frac{b^2}{4}$.

Answer: (c)

8. What is the least positive value of k for which $2x^2 + kx - 4 = 0$ has rational roots? (a) $\pm 2\sqrt{2}$ (b) 2 (c) ± 2 (d) $\sqrt{2}$ **CBSE 2023, 1 mark**

SOLUTION

The roots are rational when the discriminant is a perfect square. $D = k^2 - 4(2)(-4) = k^2 + 32$.

Among the options, $k = \sqrt{2}$ gives $D = 34$, which is not a perfect square. $k = 2$ gives $D = 36 = 6^2$.

Answer: (b)

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